2 - LOGICAL AND ALGORITHMIC THINKING (Part 2)

True/False:

1. Logic and algorithms are the same thing. F
2. Algorithms can make logical decisions. ✓
3. Logic is sufficient for building functioning systems. F
4. An algorithm is a sequence of clearly defined steps. ✓
5. Algorithms are only useful for computers. F
6. Humans have no intuitive understanding of algorithms. F
7. The definition of an algorithm has been around for centuries. ✓ F
8. Algorithms operate on data, not physical objects. ✓
9. Every step in an algorithm must have a precise meaning. ✓
10. The order of steps in an algorithm matters. ✓
11. State in an algorithm refers to the current values of the things the algorithm is keeping track of. ✓
12. There is a global view when it comes to algorithms. F
13. Once a step in an algorithm is executed, the computer remembers it. F
14. Variables are used in algorithms to remember things from previous steps. ✓

Multiple Choice:

1. ***What is the relationship between logic and algorithms?***

   a) They are identical.

   b) Algorithms build on logic.

   c) Logic builds on algorithms.

   d) They are completely unrelated.

2. ***What is the primary function of an algorithm?***

   a) To provide a set of logical rules.

   b) To integrate logical rules and execute actions.

   c) To replace logical thinking.

   d) To confuse humans and machines.

3. ***Which of the following is NOT a characteristic of an algorithm?***

   a) A sequence of steps

   b) Ambiguous instructions

   c) Clear start and end points

   d) Finite set of instructions

4. ***What does "definiteness" mean in the context of algorithms?***

a) The algorithm must be complex.

b) Each step must have a precise meaning.

c) The algorithm must be short.

d) The steps can be ambiguous.

5. ***Why is the sequence of steps important in an algorithm?***

a) It doesn't matter.

b) It determines the outcome of the algorithm.

c) It makes the algorithm easier to read.

d) It only matters for computer algorithms.

6. ***What is "state" in an algorithm?***

a) The location of the computer.

b) The current values of the data the algorithm is using.

c) The number of steps in the algorithm.

d) The programming language used.

7. ***What is the implication of "no global view" in algorithms?***

a) The algorithm can see everything at once.

b) The algorithm only sees the current state.

c) The algorithm has a long-term memory.

d) The algorithm can predict the future.

8. ***How do algorithms "remember" things from previous steps?***

a) They have a built-in memory.

b) They use variables.

c) They look back at previous steps.

d) They don't need to remember anything.

True/False:

1. Iteration is another name for looping.
2. Iteration requires writing out each step of a repeated process.
3. A variable cannot be used to control the execution of an algorithm.
4. Selection in algorithms is also known as a conditional.
5. Conditions can only be used to control loops.
6. Pseudocode is a formal language that computers can execute directly.
7. Pseudocode can be helpful for translating ideas into real program code.
8. Computers have innate intelligence and common sense.
9. Computers will try to interpret your instructions in different ways.
10. Ambiguity in algorithms is acceptable as long as the computer can guess what you mean.
11. "And" can sometimes imply sequencing in natural language.
12. "Everyone whose surname begins with A and B" is a logically sound statement.

Multiple Choice:

1. ***What is the purpose of iteration in algorithms?***

   a) To confuse the computer

   b) To repeat a series of steps

   c) To execute each step only once

   d) To make the algorithm more complex

2. ***What is a conditional (selection) used for in algorithms?***

   a) To repeat steps a specific number of times

   b) To test a variable's value and make a decision

   c) To store data

   d) To confuse the programmer

3. ***What is pseudocode?***

   a) A formal programming language

   b) An informal way of writing an algorithm

   c) A type of computer hardware

   d) A way to confuse humans

4. ***What is a common "gotcha" when working with algorithms?***

   a) Computers are very forgiving of ambiguity.

   b) Computers have common sense.

   c) Computers do exactly what they are told, even if it's wrong.

   d) Computers can understand natural language perfectly.

5. ***Why is clarity and precision so important in algorithms?***

   a) Because computers are intelligent and can figure out what you mean.

   b) Because computers execute instructions literally.

   c) Because it makes the code look nicer.

   d) Because it makes the algorithm run faster.

6. ***What is the problem with the statement "Everyone whose surname begins with A and B"?***

   a) It's too long.

   b) It's logically impossible.

   c) It's grammatically incorrect.

   d) It's too complex.

7. ***What is the best way to avoid confusion with logical operators?***

   a) Use natural language freely.

   b) Rely on intuition.

   c) Always use the correct operators and truth tables.

   d) Avoid using logical operators altogether.

Critical Thinking Questions

1. In what real-world scenarios (outside of programming) might the order of precedence affect decision-making?
2. The text mentions that mistakes in logic can lead to incorrect results. Can you think of an example from daily life where a misunderstanding of logic caused an issue?
3. If you were designing a grading system, how would you ensure it correctly assigns grades without logical errors?
4. Why do you think natural language can sometimes be misleading when expressing logical conditions?
5. How does understanding logical operators and precedence help in solving complex problems, even beyond computing?
6. Could there be situations where intentionally altering the order of precedence might be useful? If so, give an example.
7. The text suggests using proper logical notation from the start to avoid errors. Do you think this principle applies to other fields, such as law, business, or science? Why or why not?